

Defense Evasion Investigation Write-Up

Byte Doctor Reyes is investigating a stealthy post-breach attack where several expected security logs and Windows Defender alerts appear to be missing. The investigation focused on identifying defense evasion techniques used by the attacker after compromising the system.

Enumeration

The investigation began by analyzing Windows event logs, PowerShell logs, and Sysmon process creation events from the compromised host. The objective was to identify suspicious activity related to defense evasion and missing security logs.

During the analysis, several suspicious PowerShell commands and registry modifications were discovered. The attacker appeared to intentionally disable or bypass multiple Windows security mechanisms.

Key findings during enumeration included:

- Modification of the LSA protection registry key
- Windows Defender configuration changes
- AMSI bypass activity
- Safe Mode boot configuration changes
- PowerShell history logging suppression

Vulnerability Identification

The attacker abused legitimate Windows administrative tools and PowerShell functionality to weaken endpoint protection and reduce forensic visibility.

Question 1:

The attacker disabled LSA protection by modifying the following registry key:
HKLM\SYSTEM\CurrentControlSet\Control\LSA

Question 2:

The first PowerShell command used to disable Windows Defender was:
Set-MpPreference -DisableIOAVProtection \$true -DisableEmailScanning \$true
-DisableBlockAtFirstSeen \$true

Question 3:

The AMSI bypass script patched the following DLL function:
AmsiScanBuffer

Question 4:

The attacker configured the machine to reboot into Safe Mode using:
bcdedit.exe /set safeboot network

Question 5:

The attacker disabled PowerShell command history logging with:
Set-PSReadlineOption -HistorySaveStyle SaveNothing

Exploitation

The attacker successfully performed multiple defense evasion techniques after gaining access to the compromised system. The sequence of events demonstrated an intentional effort to reduce detection capabilities and hide malicious activity.

The likely attack chain was:

1. Disable LSA protection
2. Disable multiple Windows Defender protections
3. Patch AMSI in memory
4. Configure Safe Mode boot settings
5. Disable PowerShell command history logging

These actions allowed the attacker to operate stealthily while minimizing evidence available to defenders and investigators.